

Eric Xie

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Education

University of Waterloo

Bachelor's of Computer Engineering, Co-op

Courses: Data Structures and Algorithms, Machine Learning, Databases, Networks, Concurrency

Expected Apr. 2027

Waterloo, ON

Experience

Amazon

May. 2026 – Aug. 2026

Software Engineer Intern

Vancouver, BC

- Built a data lineage system using **AWS Step Functions** and **Lambda** to trace **300+** tax fields across **80+** downstream services, reducing schema impact analysis from **2 weeks** to **~4 hours**
- Implemented an LLM classification pipeline using **Bedrock** with progressive context escalation to determine how fields are consumed and propagated across downstream services, confidently resolving **94.5%** of usages
- Architected a parallel orchestration layer using nested state machines to recursively trace field propagation, enabling concurrent execution with branch-level fault isolation

Gem

Jan. 2026 – Apr. 2026

Software Engineer Intern

San Francisco, CA

- Developed a GDPR data retention system to anonymize expired records via daily jobs, processing **6.5M+** profiles
- Built a job activity feed tracking **18 event types** with field-level diffs, logging **10k+** daily events across **1,600+** teams
- Designed a hook system to centralize job activity logging, using **12 GraphQL DataLoaders** to eliminate **N+1 queries**
- Optimized Talent Pipeline queries across ATSes, reducing daily timeouts to **<20** and improving P90 load time by **30%**

Shopify

Jan. 2025 – Apr. 2025

Software Engineer Intern

Toronto, ON

- Engineered webhooks for **2M+** merchants using **Ruby on Rails** and **Kafka** to update Milestones awards status
- Rebuilt the Shopify Ventures website by migrating to **React**, **Hydrogen**, and **GraphQL**, improving load time by **50%**
- Reduced SQL injection attempts in supply store by **98%** through input sanitization and parameterized queries
- Improved perceived performance of cart checkout experience by **25%** by implementing optimistic rendering

Lawbrokr

May 2024 – Aug. 2024

Software Developer Intern

Toronto, ON

- Developed a Google Ads manager optimizing **\$50k+** monthly spend using **GPT-4** for automated keyword generation
- Built an image processing pipeline using **GCP** and **Cloud CDN** with automatic compression for **15k+** daily requests
- Led the development of a website builder used by **95+** law firms to create landing pages with **Rails** and **PostgreSQL**
- Optimized page rendering to reduce LCP delay by **40%** through lazy-loading using **Turbo Frames** and **Tailwind CSS**

BMO Financial Group

Sep. 2023 – Dec. 2023

Software Engineer Intern

Toronto, ON

- Designed and built an **S3** cleanup system handling **100k+** daily deletions using **AWS Lambda** and **PostgreSQL**
- Implemented AES encryption for **1M+** sensitive customer records using **Python** and **AWS KMS**
- Optimized data fetching speeds by **50%** for OCR document tagging by implementing **Flask** in-memory caching
- Reduced pre-release testing time by **80%** by automating unit tests using **Bash** and **pytest** with **90%** code coverage

Leadership

Hack the North

Mar. 2024 – Oct. 2024

Frontend Organizer

Waterloo, ON

- Organized Canada's largest student-run hackathon hosting **1500+** hackers annually from **70+** countries
- Developed hackthenorth.com using **React** and **Styled Components**, attracting **250k+** monthly visitors
- Built a responsive hacker application tool used to submit **10,000+** applications using **GraphQL** and **TypeScript**
- Led a volunteer management tool used by **500+** volunteers to schedule shifts and check-in attendees

Skills

Languages: Python, C/C++, Java, JavaScript/TypeScript, Ruby, SQL, Bash

Frameworks/Tools: AWS, React, GraphQL, Docker, Redis, Rails, GCP, PostgreSQL, MongoDB